

Benchmarking Large Language Models in Evidence-Based Medicine

Jin Li , Yiyan Deng , Qi Sun , Junjie Zhu , Yu Tian , Jingsong Li , and Tingting Zhu 

Abstract—Evidence-based medicine (EBM) represents a paradigm of providing patient care grounded in the most current and rigorously evaluated research. Recent advances in large language models (LLMs) offer a potential solution to transform EBM by automating labor-intensive tasks and thereby improving the efficiency of clinical decision-making. This study explores integrating LLMs into the key stages in EBM, evaluating their ability across evidence retrieval (PICO extraction, biomedical question answering), synthesis (summarizing randomized controlled trials), and dissemination (medical text simplification). We conducted a comparative analysis of seven LLMs, including both proprietary and open-source models, as well as those fine-tuned on medical corpora. Specifically, we benchmarked the performance of various LLMs on each EBM task under zero-shot settings as baselines, and employed prompting techniques, including in-context learning, chain-of-thought reasoning, and knowledge-guided prompting to enhance their capabilities. Our extensive experiments revealed the strengths of LLMs, such as remarkable understanding capabilities even in zero-shot settings, strong summarization skills, and effective knowledge transfer via prompting. Promoting strategies such as knowledge-guided prompting proved highly effective (e.g., improving the performance of GPT-4 by 13.10% over zero-shot in PICO extraction). However, the experiments also showed limitations, with LLM performance falling well below state-of-the-art baselines like PubMedBERT in handling named

entity recognition tasks. Moreover, human evaluation revealed persisting challenges with factual inconsistencies and domain inaccuracies, underscoring the need for rigorous quality control before clinical application. This study provides insights into enhancing EBM using LLMs while highlighting critical areas for further research.

Index Terms—Large language models, evidence-based medicine, clinical NLP, Prompt engineering.

I. INTRODUCTION

EVIDENCE-BASED medicine (EBM) is a systematic approach to clinical decision-making that integrates the best available evidence from research studies with individual clinical expertise, as well as patient values and preferences [1]. It requires effective identification, evaluation, and application of the most robust scientific evidence, with a preference for the findings of randomized controlled trials (RCTs), systematic reviews, and meta-analyses [2]. EBM represents the paradigm shift in modern healthcare towards providing high-quality, data-driven care that leads to improved patient outcomes. The adoption of an evidence-based approach in clinical settings is crucial, as it promotes the consistent use of rigorously evaluated treatments through empirical research, while also incorporating patient values and preferences to ensure care is tailored to individual needs, thereby advancing precision medicine [3]. By synthesizing current best evidence, integrating clinical expertise and patient preferences, it enables healthcare professionals to select therapeutic options that are scientifically valid, while also aligned with individual patient needs and circumstances [4].

However, practising EBM rigorously has become an immense challenge due to the explosive growth of biomedical literature across diverse sources like clinical trials, observational studies, and even social media platforms. The sheer volume and free-text nature of this heterogeneous evidence base compounds the difficulty of systematically retrieving, evaluating, and synthesizing the most relevant and highest-quality information [5]. Specifically, EBM requires precise inclusion and exclusion criteria when aggregating the evidence. The widely-adopted PICO (patient, intervention, comparison, outcome) framework illustrates the granularity at which randomized controlled trials must be filtered to formulate specific clinical queries and conduct systematic reviews [6]. Given the potential for harmful consequences from mistakes in evidence-based clinical practice, this comprehensive evidence appraisal must be conducted meticulously and accurately.

Received 26 May 2024; revised 6 September 2024 and 20 September 2024; accepted 13 October 2024. Date of publication 21 October 2024; date of current version 9 September 2025. This work was supported in part by the National Natural Science Foundation of China (NSFC) under Grant 82302352 and Grant 82172069, and in part by the Fundamental Research Funds for the Central Universities. (Corresponding author: Jingsong Li.)

Jin Li, Yiyan Deng, and Junjie Zhu are with the Institute for AI in Medicine, School of Artificial Intelligence, Nanjing University of Information Science and Technology, Nanjing 210044, China.

Qi Sun is with the Department of Pathology, Nanjing Drum Tower Hospital, The Affiliated Hospital of Nanjing University Medical School, Nanjing 210008, China.

Yu Tian is with the Engineering Research Center of EMR and Intelligent Expert System, Ministry of Education, College of Biomedical Engineering and Instrument Science, Zhejiang University, Hangzhou 310007, China.

Jingsong Li is with the Engineering Research Center of EMR and Intelligent Expert System, Ministry of Education, College of Biomedical Engineering and Instrument Science, Zhejiang University, Hangzhou 310007, China, and also with the Research Center for Data Hub and Security, Zhejiang Lab, Hangzhou 311121, China (e-mail: ljs@zju.edu.cn).

Tingting Zhu is with the Institute of Biomedical Engineering, Department of Engineering Science, University of Oxford, OX3 7DQ Oxford, U.K.

The code is publicly available on Github <https://github.com/jli0117/EBM-LLMs>.

Digital Object Identifier 10.1109/JBHI.2024.3483816



Despite the state-of-the-art performance of LLMs on many benchmark tasks, there are valid concerns around their tendency to hallucinate or generate factually inconsistent outputs, especially when operating outside of their training distributions [17]. Recent research has highlighted instances where LLMs make overly confident claims misaligned with truth, or struggle to identify salient information when summarizing long contexts. Such hallucinations and factual errors represent a serious pitfall if LLMs are employed for high-stakes tasks like synthesizing medical evidence for EBM without rigorous verification. As such, thorough human evaluation becomes

Therefore, this study aims to comprehensively evaluate the performance of LLMs across the key EBM stages [13], including *evidence retrieval*, *evidence synthesis*, and *evidence dissemination*, as depicted in Fig. 1 of the proposed evaluation framework. Our study aims to leverage LLMs to meet the rigorous demands of evidence-based practices, bridging the gap between their theoretical promise and practical utility in healthcare. Our primary contributions include:

- We conduct extensive comparisons of seven LLM models, including proprietary models like GPT-4 and open-source models like `Flan-UL2` [18], on five EBM datasets covering the key stages in EBM. These essential tasks encompass fundamental NLP capabilities such as named entity recognition (NER), question answering (QA), and text generation.
- With zero-shot performance as baselines, we explore and compare the effectiveness of prompting strategies that leverage medical domain knowledge (e.g., providing PICO frameworks), as well as ICL and Chain-of-Thought (CoT) approaches to adapt LLMs to specific EBM tasks. Traditional supervised state-of-the-art models in clinical NLP tasks such as PubMedBERT [19] are also finetuned and evaluated as baselines.
- To ensure safe deployment for EBM tasks involving open-ended text generation, we conduct rigorous human evaluation focused on key criteria such as factual consistency, fabricated information, and hallucination, for the evidence synthesis and evidence dissemination stages.

A. NLP Techniques for EBM

NLP has become indispensable in EBM by automating its labor-intensive processes. For evidence retrieval, Bidirectional

Encoder Representations from Transformers (BERT) has become essential for processing complex clinical texts, significantly improving the extraction of relevant clinical entities and relationships. MT-Clinical BERT [20] showcases the efficiency of multi-task learning by handling multiple clinical tasks simultaneously. Additionally, DocR-BERT [21] leverages cross-attention and Gaussian distributions to extract relations from biomedical texts. Recent research has focused on improving the granularity and precision of information extraction from clinical texts. Hu et al. [22] proposed a section-specific approach for identifying PICO elements in RCT abstracts. Zhang et al. [23] introduced a span-based model for managing overlapping entities in biomedical literature.

For RCT summarization, Bidirectional and Auto-Regressive Transformers (BART) [24] excels in creating concise summaries but may include inaccuracies, as shown by Wallace et al. [25]. In biomedical QA, models like BioBERT [26], PubMedBERT [19], and BioELECTRA [27], pretrained on specialized biomedical corpora, have significantly improved performance. For evidence dissemination, Guo et al. [28] showed that BART-generated biomedical summaries achieved a ROUGE-L score of 50.24 and a Flesch-Kincaid readability score of 13.30, demonstrating quality comparable to expert-created lay summaries. Furthermore, CO-Search [29] integrates multiple EBM stages by combining semantic search, question answering, and abstractive summarization to retrieve and synthesize COVID-19 literature. Despite these advancements, there remains a gap in comprehensively evaluating advanced NLP models across the entire EBM workflow—spanning evidence retrieval, synthesis, and dissemination—with a structured comparison for clinical application.

B. LLMs in Clinical Applications

LLMs have quickly become valuable tools in clinical applications, demonstrating versatility across various tasks. Research shows LLMs can summarize complex medical texts, aiding quick information assimilation Tang et al. [12], and extract critical data from free-text radiology reports, streamlining documentation and decision-making Le et al. [30]. They can also synthesize high-dimensional electronic health records, with implications for predictive analytics and personalized medicine Theodorou et al. [9]. MedPodGPT, a multilingual LLM augmented with medical audio podcasts, has shown effectiveness in medical QA tasks across multiple languages Jia et al. [31]. Katz et al. [11] found GPT-4 performed comparably to or better than resident physicians in fields like psychiatry and surgery.

However, challenges remain, such as inaccuracies from general-purpose LLMs, which can be harmful in medical contexts [8]. Fine-tuning LLMs on medical-specific data, like MedAlpaca [32], aims to improve accuracy. Additionally, the issue of hallucination—where LLMs generate confident but incorrect information—highlights the need for careful evaluation by clinicians to ensure patient safety [10].

Therefore, thorough evaluations are needed to assess and compare various LLMs, including both general-purpose LLMs

like GPT-4 and fine-tuned models such as MedAlpaca in clinical settings. Our work also fills the gap by conducting extensive human evaluations of LLM outputs in EBM tasks, focusing on factual consistency and clinical utility.

C. Prompt Engineering for Enhancing LLM Performance

Prompt engineering offers an efficient alternative to extensive fine-tuning, enabling LLMs to adapt to specific tasks through techniques like ICL and CoT prompting. ICL guides model predictions by providing a few examples during inference, as shown by Brown et al. [33] with GPT-3's few-shot learning capabilities. Agrawal et al. [34] demonstrated competitive results in clinical information extraction with minimal data used. Van et al. [35] highlighted ICL's effectiveness in improving clinical text summarization, applicable to diverse tasks such as radiology and doctor-patient dialogues. CoT prompting, introduced by Wei et al. [36], involves reasoning through intermediate steps, enhancing performance in complex NLP tasks such as arithmetic reasoning. Li et al. [37] further combined it with retrieval mechanisms to improve few-shot medical NER.

In addition to ICL and CoT prompting, other techniques have enhanced LLMs in clinical settings. Hu et al. [38] improved accuracy by using error analysis-based instructions and contextual prompts based on annotation guidelines, specifically for clinical NER tasks. Chuang et al. [39] introduced Soft Prompt-Based Calibration (SPeC), which calibrates LLMs to reduce variability across models, enhancing reliability in clinical note summarization. Our study builds on this by evaluating LLMs with various prompting engineering techniques such as ICL, CoT prompting. Our study also introduces a new strategy called knowledge-guided prompting, which is closely related to the work of Hu et al. [38], but goes further by exploring a broader range of scenarios across different stages of EBM and providing deeper insights into its impact throughout these stages.

III. METHOD AND MATERIALS

A. Task and Datasets

To evaluate the performance of different LLMs, we consider datasets from four tasks that are part of the pipeline of *evidence retrieval*, *evidence synthesis*, and *evidence dissemination* in EBM.

1) *PICO Extraction*: PICO extraction is the task of identifying the key components of a clinical study from its text, namely the Patient (e.g., patients with fibrocystic breast disease), Intervention (e.g., Dihydroergocryptine tablets), Comparison or Control (e.g., placebo), and Outcome (e.g., presence of nodules). By accurately extracting these PICO elements from RCT reports, researchers and clinicians can quickly assess the relevance of a study to their specific clinical question or patient population. We evaluated LLMs on the EBM-NLP corpus released by Nye et al. [42]. EBM-NLP contains 4,993 annotated PubMed abstracts on clinical trials, with each abstract annotated with entities in the PICO framework, as shown in Fig. 1.

TABLE I
DATASET USED IN THIS STUDY

Task	Dataset	EBM Pipeline	NLP Task	Train/Dev/Test	Avg. Tokens
PICO extraction	EBM-NLP [22]	Evidence Retrieval	Named Entity Recognition	300/50/150	102
Biomedical QA	BioASQ-7b [19], PubMedQA [40]	Evidence Synthesis	Question Answering	670/85/140, 450/50/500	390*, 213*
RCT summarization	Wallace et al. [25]	Evidence Synthesis	Text Summarization	1734/215/194	799
Medical text simplification	Medical-Paper Blog dataset [41]	Evidence Dissemination	Text Simplification	420/140/140	23

* This includes both the context (reference text) and the question.

The original EBM-NLP corpus has certain limitations, including the lack of expert annotation and complex annotation schema leading to inconsistencies. Therefore, we adopt the version re-annotated by Hu et al. [22]. This is a randomly selected and re-annotated dataset of 500 RCT abstracts from the EBM-NLP corpus that addresses these drawbacks. In order to enhance the quality and reliability of the dataset, the re-annotation adhered to clear guidelines to maintain consistency and involved medical experts in the annotation process. The Inside-Outside-Beginning (IOB) format is used during the annotation of the PICO elements at the word level for EBM-NLP dataset. In IOB, each token is labeled to indicate whether it is outside an entity (O), the beginning of an entity (B), or inside an entity (I). This allows the data to specify the text span for each PICO element. In this study, the training, validation, and test sets are further divided into 300, 50, and 150, respectively.

2) Biomedical QA: The biomedical QA task represents an important challenge in biomedical natural language understanding (NLU) and is crucial for EBM as part of the evidence synthesis process. As such, this task serves as an effective benchmark for evaluating the biomedical knowledge and reasoning capabilities of LLMs in QA systems. In this study, we use two well-established datasets for biomedical QA evaluation: the BioASQ (Biomedical Semantic Indexing and Question Answering) dataset [43] and the PubMedQA dataset [40].

The BioASQ dataset contains multiple QA tasks annotated by biomedical experts. In this study, we focus specifically on the yes/no QA task (Task 7b), and utilize the version that has been included as a benchmark in BLURB (Biomedical Language Understanding & Reasoning Benchmark) [19]. They cover various biomedical topics in clinical studies, including therapies, biological mechanisms, and relationships between genes, proteins, diseases, and cellular processes. The split for training, development, and testing comprised 670, 85, and 140 questions, respectively. Additionally, we incorporated the PubMedQA dataset, a novel biomedical QA dataset derived from PubMed abstracts. We used the expert-annotated version of the dataset, containing 1k instances. This dataset covers a wide range of medical topics, including various study designs and healthcare subjects like treatment outcomes, prognosis, and risk factors. Compared to BioASQ Task 7b, PubMedQA presents a greater challenge, requiring models to generate both short answers (yes/no/maybe) and comprehensive long-form answers (summarization of the conclusion), demanding deeper reasoning and summarization capabilities. The dataset was split into training, development, and testing sets with 450, 50, and 500 questions, respectively.

3) RCTs Summarization: Generating high-quality narrative summaries that synthesize biomedical evidence from studies is

crucial for enabling EBM. These summaries allow practitioners to quickly understand the consolidated findings, strengths, limitations, and implications across all relevant clinical trials on a given topic. Biomedical text summarization can be categorized into extractive and abstractive approaches [44]. Extractive summarization involves selecting and concatenating the most salient sentences or passages from the input texts to form the summary. In contrast, abstractive summarization aims to generate novel sentences that capture the crucial information concisely, and is generally considered a more challenging task. In this study, we evaluate the performance of LLMs in multi-document abstractive summarization, which aims to produce concise narrative summaries from multiple published studies of relevant RCTs.

We adopt the dataset proposed by Wallace et al. [25] for evaluation, which comprises 4,528 Cochrane systematic reviews. The input to LLMs consists of the titles and abstracts of the RCTs included in a given Cochrane systematic review. The gold standard target is the *Authors' Conclusions* section from the abstract of that Cochrane review. To account for the limited context window in LLMs, we cap the maximum number of RCTs for a single review at 5, resulting in 1734, 215, and 194 reviews for training, validation, and testing, respectively.

4) Medical Text Simplification: Medical text simplification refers to the process of simplifying complex medical text to make it more understandable, especially for patients and caregivers with limited health literacy. A large portion of the online medical information is written for expert audiences using technical jargon, which creates barriers to comprehension by non-experts. Solving the challenge of medical text simplification can facilitate EBM by improving the dissemination and understanding of health information among patients, aiding in informed decision-making.

Medical text simplification generally comprises three main subtasks, including lexical simplification, syntactic simplification, and sentence compression [41]. Lexical simplification replaces complex medical terminology with semantically equivalent plain language explanations (e.g., *hyperlipidemia* → *high cholesterol*). Syntactic simplification reduces grammatical complexity while retaining the meaning (e.g., *the patient has been prescribed a regimen of daily administration* → *the patient took medicine every day*). Sentence compression produces a summary by removing non-essential information.

To evaluate the performance of LLMs on the medical text simplification task, we utilized the Medical Paper-Blog Dataset [41]. This dataset comprises 706 aligned pairs of complex and simplified sentences from the medical domain. The complex sentences were sourced from research papers, while the simplifications were obtained from corresponding scientific blogs. Importantly,

TABLE II
TASK-SPECIFIC PROMPT USED FOR QUERYING MODELS

Task	Prompts
PICO extraction	Task Description You are a skilled medical expert. Your task is to generate an HTML version of an input text, marking up specific entities related to healthcare. The entities to be identified are: "Participant", "Intervention", "Control", and "Outcomes". Use HTML tags to highlight these entities. Each should have a class attribute indicating the type of the entity. Markup Format Use to denote a participant entity. Use to denote an intervention entity. Use to denote a control entity. Use to denote an outcome entity. Leave the text as it is if no such entities are found. Entity Recognition Guide [entity recognition guide for PICO extraction task] Examples (N-shot) [i=1,2,...N] Input i: <example.input.text.i> Annotated output i: <example.output.i> Input Text <input.text>
Biomedical QA	Task Description You are a skilled medical expert. Considering the information from a biomedical study provided in the reference text, is it correct to conclude that "<question>"? Please respond with "yes" or "no". Please first respond with "yes" or "no", followed by a brief explanation of your reasoning process, ensuring that your explanation aligns with the study's findings. Examples (N-shot) [i=1,2,...N] Question i: <example.question.i> Reference text i: <example.reference.text.i> Answer i: <example.answer.i> Input Text Question: <question> Reference text: <reference>
RCT summarization	Task Description You are a skilled medical expert. Consolidate the information from these randomized controlled trial abstracts into a comprehensive summary. Summarization Guide [summarization guide for RCT summarization task] Examples (N-shot) [i=1,2,...N][j=1,2,... # References in Example i] Reference Text <j> in Example <i> Title: <example.title.i.j> Abstract: <example.abstract.i.j> Summarization <example.summarization.i> Input Text [j=1,2,... # References in Input] Reference Text <j> Title: <title.j> Abstract: <abstract.j>
Medical text simplification	Task Description You are a skilled medical expert. Simplify the given medical text, making complex medical information accessible and relevant to people without a medical background. Summarization Guide [summarization guide for medical text simplification task] Examples (N-shot) [i=1,2,...N] Input i: <example.input.text.i> Simplified output i: <example.output.i> Input Text <input.text>

Blue text denotes domain-specific prefixes. Yellow, blue and green highlights represent ICL demonstrations, knowledge-guided prompting, and CoT prompting, respectively. Input variables are denoted by red text enclosed in <>.

the sentence pair alignments were manually validated by medical experts to ensure high quality and eliminate errors. The dataset is divided into 420 instances for training, 140 for validation, and 140 for testing in our study.

B. Prompting

In this section, we explore three key prompting strategies — ICL, knowledge-guided prompting, and CoT — that are commonly used in biomedical NLP tasks. The following subsections provide detailed implementations of these prompting methods as applied to various EBM tasks in this study.

1) Basic Prompt: To effectively leverage LLMs for the diverse EBM tasks, we designed tailored prompts to provide relevant instructions, guiding the models toward desired outputs. We structured our basic prompts with prefixes, task descriptions, and input text. The task-specific instructions are listed in Table II. In constructing these prompts, we used markdown format to ensure consistency and standardization across different models, as this format is widely recognized for structuring prompts effectively [38]. The basic prompt consists of an instruction and user input, which forms the foundational approach to guiding LLMs. As shown in Table II, the task description section provides specific instructions that the LLM is expected to follow

TABLE III
MODEL CARDS OF LLMs USED IN THIS STUDY

Model	Size	Context window	Open-source	Type
gpt-3.5-turbo-instruct	Unknown*	4,096 tokens	✗	Autoregressive
gpt-3.5-turbo-1106	Unknown*	16,385 tokens	✗	Autoregressive
gpt-4-0613	Unknown ^o	8,192 tokens	✗	Autoregressive
Gemini-1.0-pro	Unknown	32,768 tokens	✗	Autoregressive
Flan-UL2-20b	20B	2,048 tokens	✓	Seq2seq
Med-Alpaca-7b	7B	2,048 tokens	✓	Autoregressive
PMC-LLaMA-13b	13B	2,048 tokens	✓	Autoregressive

* The exact model size of gpt-3.5-turbo series is not publicly known, but it is likely similar in scale to GPT-3 models, i.e., 175 billion parameters.

^o OpenAI has not yet disclosed the exact model size of GPT-4, but it is expected to be significantly larger than GPT-3, potentially surpassing the trillion parameter mark.

for the given EBM task. The input data presents the instance or contextual information that the LLM should process and analyze to generate a relevant output. In line with previous work showing the benefits of priming LLMs with relevant domain knowledge using prefixes, we found that incorporating prompts emphasizing medical expertise (e.g., *You are a skilled medical expert.*, annotated with blue text in Table II) resulted in better performance compared to more general prompts. The prefix sets the high-level framing for the prompt by specifying the role or expertise the LLM should adopt.

Specifically, for the PICO extraction task, in line with prior investigations that employed LLMs for clinical NER tasks [38], we instruct the models to identify and annotate the relevant PICO elements using HTML tags. This is detailed in the Markup Format section as the formatting instruction for this task. The tags are designed to indicate the specific PICO category (e.g., `<span class="outcome">patients with fibrocystic breast disease</span>`). This markup approach enables a straightforward conversion of the LLM's output into a structured format. The corresponding markup guide (i.e., the *Markup Format* part in Table II) is provided for LLMs.

We also noticed that some LLMs (including Flan-UL2 [18], Med-Alpaca [32], and PMC-LLaMA-13b [45]) are not capable of generating HTML-formatted output as expected. Therefore, for those models, we change the prompt to allow for generating free-style text, which can then be parsed and analyzed further.¹

2) ICL: ICL involves providing the LLM with a small set of annotated examples, known as demonstrations (as shown in the highlighted *Examples* section in Table II), along with the task description and the input query. We incorporate 1, 3, and 5 few-shot samples in the ICL adaptation settings, while employing zero-shot prompting as the baseline models for comparison. It is important to note that in the zero-shot setting, neither proprietary nor open-source models received any training data, ensuring a fair comparison based on their original capabilities. For the few-shot settings, all models were provided with identical sets of demonstrations for fine-tuning, maintaining consistency across comparisons.

¹The detailed prompt: In the provided reference about a clinical trial, identify words related to "Participants", "Interventions", "Control", or "Outcomes". Respond using only words or phrases found in the reference, annotating them with their corresponding entity type. The output format should be like this: "Participant entities are...", "Intervention entities are...", "Control entities are...", and "Outcome entities are...". If no relevant information is present, please respond with "None".

3) Knowledge-Guided Prompting: Knowledge-guided prompting expands upon basic prompts by integrating domain-specific expertise, addressing limitations in general-purpose LLMs when handling specialized medical tasks. This method is crucial in the medical domain, where complex biomedical concepts are essential for accurate task completion. By incorporating expert medical knowledge into prompts, this approach provides LLMs with enriched context, enabling more accurate and relevant outputs for various EBM tasks. The blue-highlighted sections as shown in Table II represent the knowledge-guided components of our prompts. By incorporating this knowledge-guided prompting approach, we aim to steer the LLMs toward more targeted and effective output generation for EBM applications. The details of the knowledge-guided prompts designed for different EBM tasks can be found in our GitHub repository.

For the PICO extraction task, the knowledge-guided prompts included definitions of each PICO element to ensure the LLMs clearly understand the concepts they are expected to identify in the task of NER. The definitions in the prompt are based on the work of Hu et al. [22], where the annotation guideline is proposed for effectively annotating all P, I, C, and O elements at a single level. For the RCT summarization task, we provided prompts to guide the LLMs in summarizing key information from multiple RCT reports, adopting protocols from prior work on effective evidence synthesis [25]. Specifically, the prompts incorporated three main guidelines: 1) Prioritizing evidence from larger studies and those with lower inferred risk of bias to favor higher quality inputs; 2) Instructing the models to focus on extracting and highlighting the crucial PICO elements, as well as prominent snippets conveying the main study findings; 3) Guiding the models to structure the summaries by synthesizing the titles and most relevant sections of the RCT abstracts. For the medical text simplification task, the prompts are designed to align with practices suggested by prior literature on generating lay-friendly summaries of medical evidence [28], [41]. This includes removing unnecessary technical details, providing relevant background, simplifying medical jargon, and restructuring complex sentences — all to enhance understandability for lay readers.

4) CoT: In this study, when employing the CoT prompting strategy for the biomedical QA task, instead of simply asking the model to predict a yes/no answer, we prompt it to walk through the logical steps of how it arrived at that conclusion based on the given biomedical reference text. As shown in Table II, the basic prompt, which is underlined (i.e., *please respond with "yes" or "no".* for the BioASQ dataset; *please first respond with "yes," "no," or "maybe," and then provide a detailed explanation in 1–2 sentences.* for the PubMedQA dataset), is enhanced by including the instruction highlighted in green: *followed by a brief explanation of your reasoning process, ensuring that your explanation aligns with the study's findings.*

IV. EXPERIMENT SETUP

A. LLMs

In this research, we assess the performance of a variety of transformer-based LLMs across multiple tasks in EBM (see

Table III). These models span a wide range of sizes, with the number of parameters varying from 7 billion to approximately 175 billion. The context window, which refers to the maximum number of input tokens a model can process, ranges from 512 tokens to 32,768 tokens. In addition to proprietary LLMs like the GPT series, we incorporate open-source LLMs such as Flan-UL2-20b [18]. We also evaluate LLMs that have been finetuned on large-scale biomedical corpora, including Med-Alpaca-7b [32] and PMC-LLaMA-13b [45]. The LLMs are classified into two categories based on their network architecture, namely sequence-to-sequence (seq2seq) models and autoregressive models [15]. The detailed descriptions for each LLM are as follows:

- *gpt-3.5-turbo-instruct*, is released in September 2023 by OpenAI. The API of *gpt-3.5-turbo-instruct* was updated from *text-davinci-003* (deprecated since January 2024). The latter is an improved version of *text-davinci-002* trained with Reinforcement Learning from Human Feedback (RLHF) during alignment.
- *gpt-3.5-turbo-1106*, released in November 2023 as part of OpenAI's GPT-3.5-turbo series,² can support an expanded context window with a maximum of 16,385 tokens. It represents the most cost-effective model within the GPT-3.5 series, while also demonstrating strong performance in traditional NLP tasks.
- *gpt-4-0613* is a significant upgrade over previous GPT models from OpenAI,³ with improved performance across a wide range of tasks, including better reasoning, multi-modal capabilities (handling text, images, and other data), and increased factual knowledge. It has been regarded as state-of-the-art on general NLP tasks and has shown impressive performance on clinical NLP tasks, such as biomedical text summarization.
- *Gemini-1.0-pro* is developed by Google,⁴ demonstrating superior performance compared to the previous flagship model, Pathways Language Model (PaLM)-2 [46], across a diverse set of evaluation tasks. The model utilizes a decoder-only Transformer architecture trained to support a 32K context length with efficient attention mechanisms, demonstrating advanced understanding and reasoning capabilities across modalities.
- *Flan-UL2-20b* is finetuned through FLAN instruction tuning on the UL2 20 billion model [18], with publicly available model checkpoints. UL2 employs a novel pre-training approach called Mixture-of-Denoisers (MoD), which combines multiple pre-training objectives into a unified self-supervised learning framework called unified language learning.
- *Med-Alpaca-7b* is a 7 billion parameter LLM fine-tuned specifically for medical domain tasks [32], based on the LLaMA (Large Language Model Meta AI) architecture [47]. Its training data is collected from various sources

such as medical Anki flashcards, and question-answer pairs generated from Wikidoc.

- *PMC-LLaMA-13b* is another LLM finetuned on LLaMA specifically for the medical domain [45]. The training data includes 4.8 million biomedical academic papers and 30,000 medical textbooks, with thorough fine-tuning to align the model with domain-specific instructions.

We finetuned domain-specific Bidirectional Encoder Representations from Transformers (BERT) models for PICO extraction and biomedical QA tasks, to establish benchmarks for traditional supervised learning methods. As a widely adopted pre-trained language model, BERT excels at capturing contextual representations from unstructured text data. Building upon BERT's architecture, specialized language models like BioBERT [26], PubMedBERT [19], and BioClinicalBERT [48] have been developed and trained on biomedical corpora, such as biomedical literature and clinical notes. These domain-adapted models have exhibited state-of-the-art performance in clinical NLP across various tasks. During the model training process, we optimized hyperparameters through a grid search on a validation set. The hyperparameter sets include learning rates of 5e-5, 2e-5, and 1e-5, batch sizes of 8, 16, and 32, and training epochs of 5, and 10. The AdamW optimizer with a weight decay of 0.01 was utilized. However, because BERT's architecture is not designed for free-text generation (as it primarily focuses on masked language modeling [24]), we did not utilize these BERT-based models for RCT summarization, medical text simplification, or generating long answers for the PubMedQA dataset in our study.

B. Evaluation Metrics

1) *Quantitative NLP Metrics*: To quantitatively evaluate the performance of LLMs in various EBM tasks, we employ NLP metrics tailored to each specific task. In the PICO extraction task, due to the frequent occurrence of multi-token PICO named entities, token-level evaluation was not employed as the primary objective is to accurately extract the complete entity spans. Instead, model performance is evaluated using entity-level precision, recall, and F1 scores. The relaxed matching criteria are adopted, where an extracted entity is considered a true positive if it overlaps with a gold entity and shares the same entity type. We followed the same evaluation script as used in the prior work by Hu et al. to ensure consistent scoring [38]. To evaluate performance on the biomedical QA task, LLMs are scored based on the accuracy, and Area Under the Receiver Operating Characteristic Curve (AUROC) of their predictions across the test set (macro-averaged). Additionally, Recall-Oriented Understudy for Gisting Evaluation (ROUGE)-L is used to evaluate long answers in the PubMedQA dataset.

For RCT summarization, we evaluate performance using commonly employed NLP metrics: ROUGE, Bilingual Evaluation Understudy (BLEU), and Metric for Evaluation of Translation with Explicit ORdering (METEOR). Higher scores indicate better matching with ground truth conclusions written by human experts. In the task of medical text simplification, we assess the quality of the LLM-generated simplifications concerning both semantic similarity and readability. We evaluated semantic

²<https://platform.openai.com/docs/models/gpt-3-5-turbo>

³<https://platform.openai.com/docs/models/gpt-4-turbo-and-gpt-4>

⁴<https://deepmind.google/technologies/gemini/pro/>

similarity using ROUGE against human references, measuring preservation of meaning from complex inputs. Readability was assessed via Flesch-Kincaid grade level [28], estimating text accessibility for different literacy levels, with lower grades indicating easier readability.

2) Human Evaluation Metrics: To comprehensively evaluate the capabilities of the LLMs in the context of EBM, we conduct a thorough human evaluation of the LLM-generated text in zero-shot settings across tasks of RCT summarization and medical text simplification. We recruit five human evaluators, including two board-certified clinicians during evaluation. For the tasks of RCT summarization, the evaluators assess the text across three key dimensions — *comprehensiveness*, *correctness*, and *coherence*. Comprehensiveness evaluates whether the generated text covers all key details and clinically relevant information from the source documents. Correctness assesses the factual accuracy of the generated text, ensuring it is consistent with and supported by the source document. Coherence rates the logical flow and cohesiveness of the generated text. For the medical text simplification task, we emphasize readability over coherence, therefore the metrics for this task are *comprehensiveness*, *correctness*, and *readability*. Readability assesses the use of plain language and clarity for a general audience while retaining essential meaning.

The evaluators provide their ratings on a 5-point Likert scale, with 1 indicating strong disagreement and 5 indicating strong agreement with regard to each dimension. To make the human evaluation feasible, we randomly sampled 20 summaries for the RCT summarization task, where the original number of RCTs for each summary ranges from 1 to 5. Similarly, for the medical text simplification task, 50 samples were randomly selected for evaluators during the assessment.

V. RESULTS

A. Quantitative Results

1) Zero-Shot Performance Comparison Across Various LLMs: We first compare the performance of different LLMs across various tasks in EBM in the zero-shot setting, which refers to the ability of a model to perform a specific task without any additional training or fine-tuning on the task-specific dataset. In the task of PICO extraction, gpt-4-0613 demonstrated the best overall performance, achieving an F1 score of 0.504 in the zero-shot setting. Med-Alpaca-7b exhibited the highest recall, with a score of 0.449. While the Med-Alpaca-7b, as well as another open-source LLM, Flan-UL2-20b demonstrated a higher precision in identifying PICO elements (highest precision score at 0.663 for Med-Alpaca-7b), they suffered from lower sensitivity, and therefore leading to a lower overall F1 score. This suggests that while high precision ensures the extraction of mostly correct entities, the models struggle to identify all instances of PICO elements, resulting in incomplete entity extraction.

In the zero-shot setting for biomedical QA tasks, Flan-UL2-20b showcased superior performance across both the BioASQ and PubMedQA datasets. On the BioASQ dataset, it achieved a remarkable accuracy of 0.950 and an AUROC of 0.952. For the PubMedQA dataset, which presents a greater challenge by

requiring models to generate both short answers and comprehensive long-form answers that demand deeper summarization capabilities, Flan-UL2-20b led with an accuracy of 0.750, an AUROC of 0.707, and the highest Rouge-L score of 0.273. The proprietary LLMs, gpt-4-0613 and Gemini-1.0-pro, also demonstrated notable performance. Conversely, the Med-Alpaca-7b and PMC-LLaMA-13b models failed to follow instructions adequately, rendering them unable to generate valid results for both short and long answers on this dataset.

For the task of summarizing RCTs, Gemini-1.0-pro achieved the highest ROUGE-1, ROUGE-L and BLEU scores. The proprietary model gpt-3.5-turbo-instruct exhibited the highest METEOR score of 0.238. In contrast to the strong performance of these proprietary models, there was a notable performance degradation observed with the open-source models, including Flan-UL2-20b, Med-Alpaca-7b, and PMC-LLaMA-13b. This indicates that their summaries had less overlap with the reference summaries, and potentially lacked the same level of comprehensiveness and factual accuracy as the proprietary models (detailed human evaluation results will be discussed in the next section). The challenges in RCT summarization arise from two key factors. First, the task's complexity extends beyond simple result summarization, requiring interpretation of intricate clinical details, statistical outcomes, and treatment effects across trials. This demands a level of medical expertise and analytical capability that current LLMs struggle to replicate. Second, the large token size of input sequences, averaging 799 tokens (see Table I), presents a significant challenge for models in processing and synthesizing information across long texts.

For the task of medical text simplification, the best performance was observed in the Flan-UL2-20b model, closely followed by LLMs finetuned on the domain-specific corpus, i.e., Med-Alpaca-7b and PMC-LLaMA-13b, both achieving relatively high ROUGE scores. Proprietary models, while performing well on readability metrics, often generate oversimplified outputs, or add background information for better comprehension that was not present in the original text. This led to lower alignment with reference labels and overall metric scores. In contrast to the RCT summarization task, the shorter token length in the simplification task potentially makes it easier for models like Flan-UL2-20b to effectively process and capture the necessary context.

2) Impact of ICL: Table IV also showcases the ICL results, with the number of demonstrations for LLMs ranging from 1, 3, to 5. Across various tasks in EBM, the adaptation strategy of ICL consistently enhanced the performance of the LLMs in almost all cases. Notably, in the PICO extraction task, the Flan-UL2-20b model's overall F1 score increased from 0.190 in the zero-shot setting to 0.247 in the one-shot setting, representing a 30% improvement even with a single example. This highlights the significant impact that ICL can have by utilizing demonstrations. However, it is important to note that more demonstrations do not necessarily lead to better results. For example, in the case of the gpt-4-0613 in the RCT summarization task, there was an increase in performance in the one-shot setting compared with the zero-shot results, but no substantial enhancement was observed when the number of demonstrations was increased

TABLE IV
PERFORMANCE COMPARISON OF LLMs IN ZERO-SHOT AND FEW-SHOT SETTINGS

	PICO extraction (EBM-NLP)			Biomedical QA (BioASQ-7b) (PubMedQA)					RCT summarization (Wallace et al. [22])					Medical text simplification (Medical-Paper Blog dataset)			
	Pre.	Rec.	F1	Acc.	AUC	Acc.	AUC	R-L	R-1	R-2	R-L	B-S	M-S	R-1	R-2	R-L	F-K
gpt-3.5-turbo-instruct (0-shot)	0.540	0.350	0.425	0.835	0.772	0.668	0.614	0.198	0.206	0.037	0.187	0.049	0.238	0.349	0.128	0.300	10.74
gpt-3.5-turbo-instruct (1-shot)	0.584	0.298	0.394	0.857	0.799	0.734	0.667	0.205	—	—	—	—	—	0.405	0.180	0.357	12.06
gpt-3.5-turbo-instruct (3-shot)	0.601	0.337	0.431	0.843	0.771	0.762	0.693	0.199	—	—	—	—	—	0.395	0.164	0.345	12.15
gpt-3.5-turbo-instruct (5-shot)	0.614	0.405	0.488	—	—	0.752	0.687	0.196	—	—	—	—	—	0.394	0.166	0.342	12.24
gpt-3.5-turbo-1106 (0-shot)	0.546	0.257	0.350	0.864	0.838	0.668	0.604	0.245	0.204	0.035	0.184	0.049	0.230	0.356	0.139	0.316	10.99
gpt-3.5-turbo-1106 (1-shot)	0.601	0.265	0.368	0.886	0.865	0.756	0.692	0.264	0.222	0.040	0.197	0.059	0.226	0.433	0.200	0.388	12.50
gpt-3.5-turbo-1106 (3-shot)	0.657	0.280	0.392	0.907	0.859	0.720	0.688	0.252	0.230	0.040	0.203	0.060	0.225	0.435	0.209	0.393	12.60
gpt-3.5-turbo-1106 (5-shot)	0.621	0.354	0.451	0.900	0.881	0.734	0.692	0.259	0.230	0.043	0.205	0.061	0.233	0.446	0.217	0.404	12.74
gpt-4-0613 (0-shot)	0.575	0.449	0.504	0.921	0.897	0.670	0.691	0.261	0.188	0.032	0.173	0.041	0.226	0.295	0.086	0.250	9.85
gpt-4-0613 (1-shot)	0.707	0.394	0.505	0.929	0.908	0.726	0.709	0.259	0.203	0.033	0.183	0.048	0.231	0.387	0.168	0.349	11.59
gpt-4-0613 (3-shot)	0.693	0.467	0.558	0.936	0.924	0.795	0.707	0.266	0.207	0.034	0.185	0.050	0.230	0.407	0.180	0.363	11.74
gpt-4-0613 (5-shot)	0.688	0.481	0.566	0.943	0.924	0.740	0.719	0.278	0.204	0.035	0.181	0.051	0.218	0.415	0.193	0.377	11.99
Gemini-1.0-pro (0-shot)	0.536	0.235	0.326	0.921	0.903	0.600	0.656	0.247	0.214	0.035	0.192	0.054	0.194	0.263	0.071	0.224	9.87
Gemini-1.0-pro (1-shot)	0.558	0.240	0.336	0.886	0.859	0.638	0.645	0.248	0.178	0.026	0.159	0.038	0.151	0.443	0.229	0.401	12.14
Gemini-1.0-pro (3-shot)	0.574	0.208	0.305	0.871	0.849	0.696	0.668	0.245	0.231	0.038	0.205	0.053	0.192	0.491	0.290	0.454	10.12
Gemini-1.0-pro (5-shot)	0.601	0.317	0.415	0.864	0.838	0.712	0.692	0.249	0.222	0.033	0.195	0.052	0.200	0.521	0.324	0.485	13.86
Flan-UL2-20b (0-shot)	0.651	0.111	0.190	0.950	0.952	0.750	0.707	0.273	0.176	0.029	0.149	0.023	0.114	0.558	0.373	0.523	14.01
Flan-UL2-20b (1-shot)	0.700	0.150	0.247	0.950	0.941	0.770	0.717	0.306	0.174	0.027	0.149	0.019	0.111	0.562	0.378	0.530	14.08
Flan-UL2-20b (3-shot)	0.679	0.224	0.337	0.979	0.978	0.760	0.712	0.299	0.179	0.031	0.153	0.022	0.114	0.564	0.377	0.529	14.41
Flan-UL2-20b (5-shot)	0.698	0.213	0.326	—	—	—	—	—	0.178	0.031	0.150	0.024	0.117	0.568	0.382	0.533	14.33
Med-Alpaca-7b (0-shot)	0.663	0.051	0.094	0.664	0.633	/	/	/	0.150	0.023	0.132	0.022	0.115	0.470	0.265	0.425	11.29
Med-Alpaca-7b (1-shot)	0.848	0.069	0.127	0.707	0.643	/	/	/	—	—	—	—	—	0.478	0.290	0.438	13.30
Med-Alpaca-7b (3-shot)	0.727	0.079	0.142	0.586	0.519	/	/	/	—	—	—	—	—	0.507	0.316	0.466	13.76
Med-Alpaca-7b (5-shot)	0.543	0.056	0.101	—	—	/	/	/	—	—	—	—	—	0.516	0.331	0.475	13.77
PMC-LLaMA-13b (0-shot)	0.357	0.028	0.052	0.664	0.733	/	/	/	0.158	0.022	0.137	0.027	0.134	0.395	0.202	0.351	12.67
PMC-LLaMA-13b (1-shot)	0.645	0.093	0.162	0.750	0.692	/	/	/	—	—	—	—	—	0.473	0.294	0.430	12.80
PMC-LLaMA-13b (3-shot)	0.622	0.066	0.119	0.629	0.524	/	/	/	—	—	—	—	—	0.483	0.296	0.442	13.17
PMC-LLaMA-13b (5-shot)	0.714	0.134	0.225	—	—	/	/	/	—	—	—	—	—	0.421	0.253	0.383	12.33

Note: "—" indicates the number of tokens exceeding the context window, "/" denote experiments were not applicable to the model.

TABLE V
COMPARISON OF ZERO-SHOT AND PROMPTED RESULTS FOR PICO EXTRACTION TASK (KNOWLEDGE-GUIDED PROMPTING) AND BIOMEDICAL QA TASK (COT), USING STATE-OF-THE-ART BERT-STYLE MODELS AS BASELINES FOR COMPARISON

		PICO extraction (EBM-NLP)			Biomedical QA (BioASQ)				Biomedical QA (PubMedQA)			
		Pre. Δ (%)	Rec. Δ (%)	F1 Δ (%)	Acc. Δ (%)	AUROC Δ (%)	Acc. Δ (%)	AUROC Δ (%)	Acc. Δ (%)	AUROC Δ (%)	R-L Δ (%)	R-L Δ (%)
Baselines	BioBERT	0.835	0.885	0.859	0.771	0.790	0.526	0.529	/	/	/	/
	BioClinicalBERT	0.774	0.864	0.816	0.693	0.662	0.550	0.533	/	/	/	/
	PubMedBERT	0.830	0.873	0.851	0.736	0.831	0.580	0.569	/	/	/	/
LLMs	gpt-3.5-turbo-instruct	0.573 $\Delta 0.345\%$	0.304 $\Delta -13.14\%$	0.397 $\Delta -6.59\%$	0.829 $\Delta -0.72\%$	0.772 $\Delta 0.00\%$	0.662 $\Delta -0.90\%$	0.661 $\Delta 7.65\%$	0.208 $\Delta 5.05\%$			
	gpt-3.5-turbo-1106	0.553 $\Delta 1.28\%$	0.269 $\Delta 4.67\%$	0.362 $\Delta 3.43\%$	0.879 $\Delta 1.74\%$	0.860 $\Delta 2.63\%$	0.690 $\Delta 3.29\%$	0.677 $\Delta 12.09\%$	0.254 $\Delta 3.67\%$			
	gpt-4-0613	0.733 $\Delta 27.48\%$	0.466 $\Delta 3.79\%$	0.570 $\Delta 13.10\%$	0.950 $\Delta 3.15\%$	0.935 $\Delta 4.24\%$	0.690 $\Delta 2.99\%$	0.696 $\Delta 0.72\%$	0.265 $\Delta 1.53\%$			
	Gemini-1.0-pro	0.562 $\Delta 4.85\%$	0.228 $\Delta -2.98\%$	0.325 $\Delta -0.31\%$	0.936 $\Delta 1.63\%$	0.924 $\Delta 2.33\%$	0.706 $\Delta 17.67\%$	0.707 $\Delta 7.77\%$	0.247 $\Delta 0.00\%$			
	Flan-UL2-20b	0.696 $\Delta 6.91\%$	0.236 $\Delta 112.61\%$	0.352 $\Delta 85.26\%$	0.957 $\Delta 0.74\%$	0.956 $\Delta 0.42\%$	0.760 $\Delta 1.33\%$	0.692 $\Delta -2.12\%$	0.281 $\Delta 2.93\%$			
	Med-Alpaca-7b	0.681 $\Delta 2.71\%$	0.046 $\Delta -9.80\%$	0.086 $\Delta -8.51\%$	0.700 $\Delta 5.42\%$	0.643 $\Delta 1.58\%$	/	/	/			
	PMC-LLaMA-13b	0.649 $\Delta 81.79\%$	0.116 $\Delta 314.28\%$	0.196 $\Delta 276.92\%$	0.693 $\Delta 4.37\%$	0.694 $\Delta -5.32\%$	/	/	/			

Note: "/" denote experiments were not applicable to the model.
The percentage of performance improvement is shown in red text, and the percentage of decrease is shown in blue text.

further. Similar trends were seen with gpt-3.5-turbo-1106 in the PubMedQA dataset: while 1-shot results showed improvement over zero-shot, subsequent increases in demonstrations (3-shot and 5-shot) did not yield significant gains, with accuracy peaking at 0.756. The first demonstration likely helped the model understand the prompts or provided an example format, but additional demonstrations offered minimal new information, resulting in only marginal performance improvements.

Moreover, the effectiveness of LLMs in leveraging demonstrations could be influenced by their context window lengths and long-sequence modeling capabilities. For example, in the biomedical QA task, models with shorter context windows, such as Med-Alpaca-7b and PMC-LLaMA-13b, exhibited limited performance gains when provided with a large number of demonstrations. On the other hand, the LLM's contextual understanding ability and instruction-following capability also limit the efficacy of the model in utilizing demonstrations. Using the GPT series models as an example, in the task of medical

text simplification, the performance improvement in the 5-shot setting was more substantial for the more recent gpt-4-0613 (50.8% increase in ROUGE-L), compared to gpt-3.5-turbo-instruct (14% increase in ROUGE-L) and gpt-3.5-turbo-1106 (27.8% increase in ROUGE-L).

3) Impact of Domain Knowledge-Guided Prompting: The domain knowledge-guided prompting strategy was implemented across the tasks of PICO extraction, RCT summarization, and medical text simplification. The results are presented in Tables V and VI respectively, with the evaluated metrics and improvement percentages against the zero-shot results calculated and annotated accordingly. For the PICO extraction task, all models demonstrate improved precision when compared to their base results. Notably, the overall F1 score shows a significant boost of at least 3% for GPT-3.5-turbo-1106, GPT-4-0613, Flan-UL2-20b, and PMC-LLaMA-13b. However, three other models experience a slight decrease in performance, ranging from 0.3% to 8.51%. As the knowledge-guided prompts for the

TABLE VI

COMPARISON OF ZERO-SHOT AND PROMPTED RESULTS FOR RCT SUMMARIZATION TASK AND MEDICAL TEXT SUMMARIZATION TASK (KNOWLEDGE-GUIDED PROMPTING)

	RCT summarization (Wallace et al. [22])					Medical text simplification (Medical-Paper Blog dataset)			
	R-1 Δ (%)	R-2 Δ (%)	R-L Δ (%)	BLEU Δ (%)	METEOR Δ (%)	R-1 Δ (%)	R-2 Δ (%)	R-L Δ (%)	F-K Δ (%)
gpt-3.5-turbo-instruct	0.193 Δ -6.31%	0.034 Δ -8.11%	0.175 Δ -6.42%	0.044 Δ -10.20%	0.232 Δ -2.52%	0.357 Δ 2.29%	0.135 Δ 5.47%	0.310 Δ 3.33%	10.73 Δ -0.09%
gpt-3.5-turbo-1106	0.209 Δ 2.45%	0.039 Δ 11.43%	0.188 Δ 2.17%	0.050 Δ 2.04%	0.236 Δ 2.61%	0.368 Δ 3.37%	0.143 Δ 2.88%	0.322 Δ 1.90%	11.10 Δ 1.00%
gpt-4-0613	0.191 Δ 1.60%	0.033 Δ 3.13%	0.173 Δ 0.00%	0.043 Δ 4.88%	0.230 Δ 1.77%	0.323 Δ 9.49%	0.105 Δ 22.09%	0.279 Δ 11.60%	11.01 Δ 11.78%
Gemini-1.0-pro	0.215 Δ 0.47%	0.034 Δ -2.86%	0.191 Δ -0.52%	0.053 Δ -1.85%	0.205 Δ 5.67%	0.277 Δ 5.32%	0.084 Δ 18.31%	0.240 Δ 7.14%	10.12 Δ 2.53%
Flan-UL2-20b	0.180 Δ 2.27%	0.032 Δ 10.34%	0.156 Δ 4.70%	0.024 Δ 4.35%	0.118 Δ 3.51%	0.557 Δ -0.18%	0.368 Δ -1.34%	0.523 Δ 0.00%	13.63 Δ -2.71%
Med-Alpaca-7b	0.160 Δ 6.67%	0.026 Δ 13.04%	0.142 Δ 7.58%	0.022 Δ 0.00%	0.115 Δ 0.00%	0.455 Δ -3.19%	0.260 Δ -1.89%	0.414 Δ -2.59%	12.83 Δ 13.64%
PMC-LLaMA-13b	0.167 Δ 5.70%	0.026 Δ 18.18%	0.148 Δ 8.03%	0.029 Δ 7.41%	0.143 Δ 6.72%	0.332 Δ -15.95%	0.154 Δ -23.76%	0.302 Δ -13.96%	11.54 Δ -8.92%

The percentage of performance improvement is shown in red text, and the percentage of decrease is shown in blue text.

PICO extraction task involved long text definitions for different entities, these models may have struggled with comprehending and mapping these lengthy textual descriptions to the respective entities. This complexity could potentially challenge the models' understanding ability, leading to incomplete annotations and lower recall scores. For the RCT summarization task, most models exhibit improvements across different metrics. In the medical text simplification task, the Flan-UL2-20b model demonstrates the best performance in terms of overlap with the reference texts. While proprietary models demonstrate performance increases using the knowledge-guided prompts, open-source models, including Flan-UL2-20b, Med-Alpaca-7b, and PMC-LLaMA-13b experience a decrease in performance. This could be due to the introduced instructions having imposed new rules or constraints that contradicted the models' original simplification strategies, consequently resulting in decreased performance.

4) *Impact of CoT Prompting*: CoT prompting strategy has been applied for biomedical QA tasks. The experimental results presented in Table V indicate that the CoT prompting strategy improved the performance of most of the LLMs compared to their respective baselines. This enhancement is especially significant in the PubMedQA dataset, where the task demands both concise answers and detailed summarizations. This improvement can be attributed to the complex nature of biomedical QA, which requires models to comprehend intricate clinical contexts and infer answers logically based on the provided information.

5) *Comparison With SOTA Clinical NLP Models*: In the PICO extraction task, despite the use of prompting strategies, there is still a significant performance gap between the BERT-style models and LLMs. To delve into the reason, NER is a sequence labeling task that requires assigning an entity-type label to each token within a sentence, while LLMs are specifically designed under a text generation paradigm [49]. This makes LLMs better suited for tasks such as open-ended generation or language modeling. As a result, the differences between the semantic labeling objective of NER tasks and the underlying architectures of LLMs lead to inferior performance when applying LLMs to resolve NER tasks like PICO extraction.

In contrast, for the biomedical QA task on BioASQ dataset, LLMs such as the GPT series, Gemini-1.0-pro, and Flan-UL2-20b in the zero-shot setting could outperform the BERT models trained under the supervised learning paradigm. The advantages are even more pronounced in the PubMedQA dataset. Additionally, BERT models, constrained by their masked language

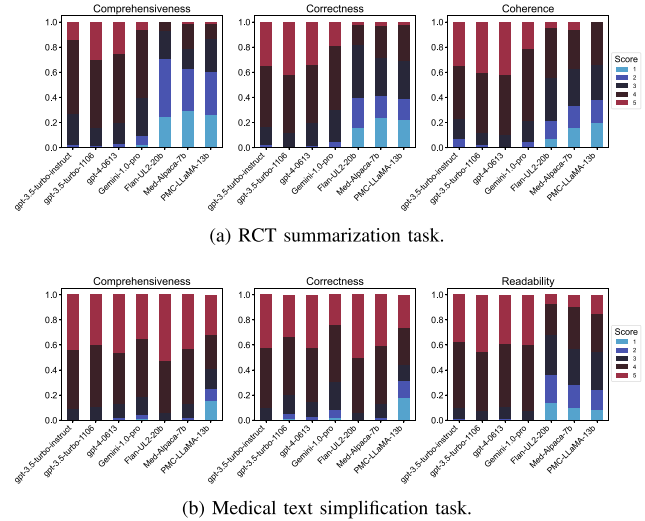


Fig. 2. Ratios of 1–5 scores in human evaluation.

modeling training, are not suited for text generation tasks, limiting our analysis to short answers. This observation highlights the remarkable contextual understanding capability of LLMs, which enables them to comprehend and reason over complex biomedical texts, even without explicit task-specific training.

B. Human Evaluation

1) *Comparing Various LLMs*: The results of human evaluation are illustrated in Figs. 2 and 3. For the RCT summarization task, it can be observed that proprietary models, including the GPT series and Gemini-1.0-pro, outperformed open-source LLMs such as Flan-UL2-20b, Med-Alpaca-7b, and PMC-LLaMA-13b. This is evidenced by the higher proportion of top scores (4 and 5) assigned to proprietary models across all three evaluation criteria. Conversely, open-source LLMs exhibited a higher ratio of low scores (1 and 2). Among the proprietary models, the GPT series achieved the highest scores from human evaluators, with comparable performance across these three models. Notably, the lower comprehensiveness scores for the open-source models suggest their generated summaries often fail to capture all the key information from the RCT abstracts. This could lead to the omission of critical details necessary for safe and effective medical practice, potentially resulting in harmful consequences for patients.

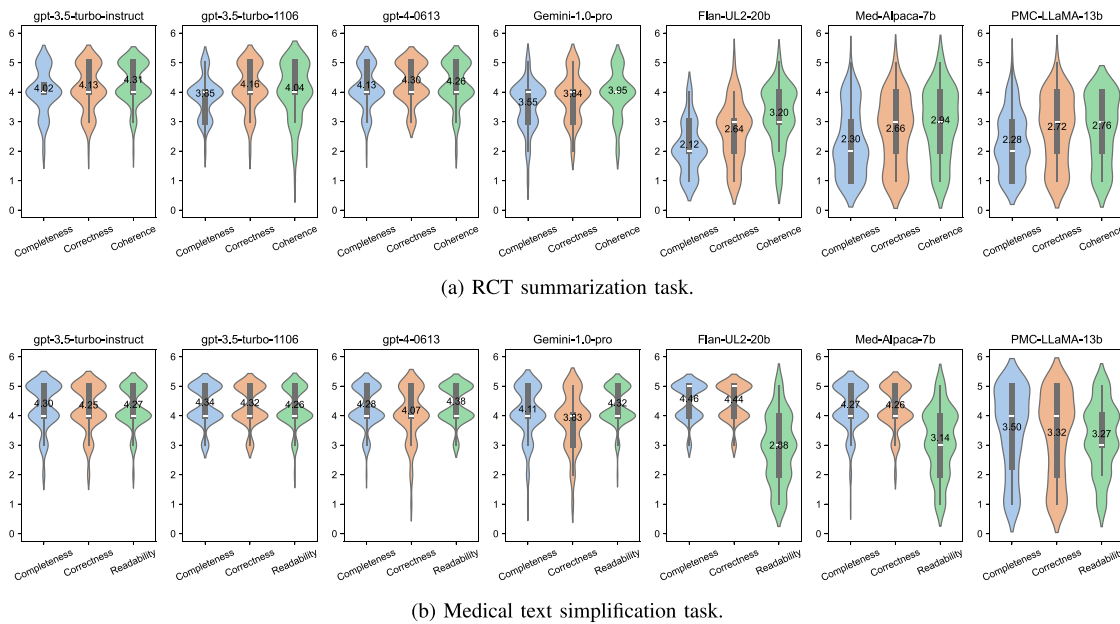


Fig. 3. Distribution of human evaluation results across three key dimensions for each LLM. The mean value for each metric is annotated for the respective model.

For the medical text summarization task, the performance across LLMs varied in terms of the evaluated dimensions. In the comprehensiveness and factual correctness dimensions, most models exhibited comparable performance, excluding PMC-LLaMA-13b, which exhibited the lowest results. The comprehensiveness scores ranged from 4.11 to 4.46 across six of the seven models (excluding PMC-LLaMA-13b). Similarly, the mean scores for correctness fell between 3.83 and 4.44, indicating a relatively small performance gap among the majority of models. However, a more significant divergence was observed in the dimension of readability. The three open-source models received lower scores in terms of readability, suggesting that these models struggle more with producing easily understandable medical text. The lower readability scores for open-source models suggest these LLMs have limited capability to fully comprehend and follow the provided instructions.

2) Hallucination: Table VII provides an overview of typical issues or errors from LLMs, categorized into several types. First of all, factual contradictions, where the LLM output contradicts the reference information, can lead to incorrect medical decisions or recommendations. For instance, Flan-UL2-20b's output contradicts the reference by stating that the incidence of atelectasis was significantly higher in patients intubated with oral tubes, which is the opposite of what the reference claims. Another typical error when summarizing all provided RCTs is evidential omissions, where the LLM fails to include all relevant information from the source RCTs. This can lead to incomplete or biased medical advice, which in turn could affect patient care and outcomes. Semantic distortions and misinterpretations can distort the intended meaning of medical information. When performing medical simplification tasks, LLMs could falsely interpret medical terminology (such as *subordinate* in the example) or the semantic meaning in the original text (such as

generation shift in the example). These errors may result in miscommunications and result in misinformed clinical judgments. Fabricated attributes and hallucinations, where the LLM generates information that is not supported by the source material, are particularly concerning in medical contexts. For example, PMC-LLaMA-13b's output hallucinates that these patients *experienced a significant improvement in their cardiac function*, which is not mentioned in the original text. Such inaccuracies, if unchecked, could cause clinical errors and potentially endanger patient safety.

VI. DISCUSSION

LLMs have demonstrated their potential as foundation models in biomedical tasks, particularly in zero-shot settings where they can effectively perform tasks like biomedical QA without task-specific fine-tuning. Their ability to reason and process complex information, even in the absence of extensive training, highlights the distinctive strengths of LLMs compared to BERT-style models that require more targeted training. However, challenges remain, particularly in tasks like PICO extraction, where LLMs struggle due to a misalignment between their text generation objectives and the semantic labeling required for NER tasks. Future research should focus on improving LLM performance in biomedical NER through architectural changes and fine-tuning [49]. Our study's findings have significant implications for personalized medicine within the framework of EBM. Improving LLM capabilities in PICO extraction and similar NER tasks is essential for gathering patient-specific data for individualized care. By achieving notable results in QA, LLMs demonstrate the potential to incorporate personalized patient information, tailoring medical advice to individual needs. Additionally, advancements in medical text simplification improve

TABLE VII
AN OVERVIEW OF ERROR TYPES

Category	Source	Example
Factual contradiction	Reference summary	Post extubation atelectasis may be more frequent after nasal intubation , particularly in very low birth weight infants. One route of intubation does not seem to be preferable to the other...
	Output summary	Flan-UL2-20b : The authors compared the incidence of post-extubation atelectasis in patients intubated with oral versus nasal endotracheal tubes. The incidence of atelectasis was significantly higher in patients intubated with oral tubes .
Certainty Illusion	Reference summary	Metronidazole, given as a single dose, is likely to provide a parasitological cure for trichomoniasis, but it is not known whether this treatment will have any effect on pregnancy outcomes...
	Output summary	gpt-3.5-turbo-1106 : In summary, the evidence from these studies suggests that treatment of asymptomatic trichomoniasis in pregnant women with metronidazole does not prevent preterm delivery, and the impact of Trichomonas infection on perinatal outcomes may be limited.
Evidence omission	Reference summary	Insufficient evidence exists from randomised trials to confirm the relative effects of open versus closed reduction of intracapsular fractures , or the effects of intra-operative impaction or compression of an intracapsular fracture treated by internal fixation.
	Output summary	Flan-UL2-20b : The results of this study suggest that closed reduction is as effective as open reduction in the treatment of displaced intracapsular fractures of the neck of the femur in young adults.
Semantic distortion	Original text	We observed unfavourable generation shifts in diabetes among men but not among women.
	Simplified text	gpt-3.5-turbo-instruct : We noticed that men were more likely to develop diabetes compared to women.
Fabricated attribute	Original text	Framingham offspring study participants who attended the seventh examination were at least 60 years old and free of dementia and stroke were included.
	Simplified text	gpt-4-0613 : The Framingham Offspring Study included participants who were at least 60 years old. These participants did not have dementia or a history of stroke .
Misinterpretation	Original text	Subordinate individuals had lower heart rates than dominants, and consequently they suffered lower rates of body mass loss during periods of restricted pellet food supply.
	Simplified text	Gemini-1.0-pro : When animals were lower in the social hierarchy , their heart rates were slower. This helped them lose less weight when food was scarce.
Hallucination	Original text	Five patients with DM2 and stage II/III HF consumed dark chocolate and a beverage containing approximately 100 mg of EPI per day for 3 months.
	Simplified text	PMC-LLaMA-13b : The patients with DM2 and stage II/III HF who consumed dark chocolate and a beverage containing approximately 100 mg of epicatechin per day for three months experienced a significant improvement in their cardiac function .

communication between clinicians and patients, empowering informed decision-making. These developments foster a patient-centred approach in EBM, emphasizing individual patient values and preferences.

We evaluated the effectiveness of various prompting strategies, including ICL, knowledge-guided prompting, and CoT reasoning, against zero-shot baselines in different EBM tasks. Firstly, ICL consistently improved LLM performance by providing relevant contextual examples, though more demonstrations do not necessarily lead to better performance. Secondly, knowledge-guided prompting improved precision and recall in domain-specific tasks like PICO extraction but was less effective in simplifying medical texts, where simpler prompts proved effective. The results also indicate that not all models benefited equally from the knowledge-guided prompts, with some open-source models such as PMC-LLaMA-13b experiencing performance decreases for certain tasks. This may stem from differences in training data or model architecture. Third, CoT prompting excelled in complex reasoning tasks like biomedical QA, significantly boosting accuracy by guiding models through structured reasoning processes. Overall, the optimal application of LLMs in EBM requires tailoring prompting strategies to specific tasks and model architectures, as a one-size-fits-all approach may not yield optimal results.

Human evaluation is essential for the responsible application of LLMs in medicine. To mitigate the potential harm of errors or hallucinations in EBM and medical practice, it is crucial to implement rigorous evaluation and validation processes when using LLMs. The results underscore that automatic metrics do not consistently correlate with human evaluations or preferences. Additionally, there are noticeable discrepancies between content

generated by LLMs and the human-authored gold standards. For instance, in the task of RCT summarization, human evaluators observed that LLMs mostly combine contents from different RCTs into a single summary, while human experts typically extract and synthesize evidence to provide nuanced inferences, reasoning, and relevant recommendations — a capability that LLMs currently lack. While LLMs show potential in supporting EBM and medical practice, their outputs must be approached with caution and rigorously scrutinized by medical specialists to ensure their reliability and appropriateness for clinical use.

There are several limitations in our current work. Firstly, although the results from employing zero-shot learning and various prompting strategies are promising, fine-tuning these models on task-specific data could potentially enhance their performance and robustness. Finetuning techniques such as Low-Rank Adaptation (LoRA) [50] have shown promise in improving model capabilities in a parameter-efficient manner. Additionally, while we focus on normal prompting strategies, incorporating retrieval-augmented generation (RAG) techniques [51] could further enhance the model's ability to access and integrate external knowledge dynamically during generation. Second, the selection of demonstration examples for the ICL approach was done randomly in the current work. Prior research has emphasized the importance of selecting high-quality demonstrations that effectively communicate the task requirements [15], as well as accounting for the variance introduced by different sets of examples, which can impact model performance and generalizability [49]. Exploring more principled approaches to demonstration selection and systematically assessing performance across various sets could lead to significant insights into model robustness and adaptability. Additionally, research has

shown that varying prompt prefixes, such as those emphasizing medical expertise (e.g., “Considering your role as a medical practitioner” [45]) or even unconventional prefixes (e.g., “You are a mystical wizard in Middle Earth” [35]), can influence LLM performance. While our study used a fixed prefix framing, we did not explore the full spectrum of possible prefixes. Future research should investigate how different prompt framings may impact task-specific outcomes. Then, catastrophic forgetting, where models forget previously learned tasks during fine-tuning, remains a potential challenge. Although not explored in this study, future research should investigate strategies to mitigate this issue and ensure stable performance across tasks during continual fine-tuning. Lastly, our human evaluation of LLMs’ outputs did not comprehensively assess their fairness and potential biases, which is critical before deploying LLMs in real-world clinical settings.

VII. CONCLUSION

The comprehensive evaluation conducted in this study sheds light on the potential of leveraging LLMs to enhance EBM practices. Our findings reveal that LLMs demonstrate impressive zero-shot capabilities in certain EBM tasks like biomedical QA, showcasing strong language understanding and reasoning skills. Moreover, our results validate the effectiveness of various prompting strategies, particularly knowledge-guided prompting, in adapting LLMs’ general language abilities to the highly specialized medical domain. However, significant challenges persist in seamlessly integrating LLMs into critical clinical workflows, such as inaccuracies in complex clinical NER task. Human evaluations further underscore that their outputs are not yet sufficiently reliable for real-world deployment without rigorous verification in place. Issues like factual inconsistencies and hallucinations pose risks in sensitive medical contexts. Overall, while challenges remain, the combination of cutting-edge language models and evidence-based practices holds tremendous promise for improving healthcare quality and patient outcomes by closely aligning clinical decision-making with the latest scientific findings.

REFERENCES

- [1] D. L. Sackett, W. M. Rosenberg, J. M. Gray, R. B. Haynes, and W. S. Richardson, “Evidence based medicine: What it is and what it isn’t,” *Bmj*, vol. 312, no. 7023, pp. 71–72, 1996.
- [2] B. Djulbegovic and G. H. Guyatt, “Progress in evidence-based medicine: A quarter century on,” *Lancet*, vol. 390, no. 10092, pp. 415–423, 2017.
- [3] S. M. Shortell, T. G. Rundall, and J. Hsu, “Improving patient care by linking evidence-based medicine and evidence-based management,” *Jama*, vol. 298, no. 6, pp. 673–676, 2007.
- [4] A. K. Akobeng, “Principles of evidence based medicine,” *Arch. Dis. Childhood*, vol. 90, no. 8, pp. 837–840, 2005.
- [5] C. Lefebvre et al., “Searching for and selecting studies,” in *Cochrane Handbook Systematic Reviews of Interventions*. Hoboken, NJ, USA: Wiley, 2019, pp. 67–107.
- [6] W. S. Richardson, M. C. Wilson, J. Nishikawa, and R. S. Hayward, “The well-built clinical question: A key to evidence-based decisions,” *ACP J. Club*, vol. 123, no. 3, pp. A12–A13, 1995.
- [7] T. Kojima, S. S. Gu, M. Reid, Y. Matsuo, and Y. Iwasawa, “Large language models are zero-shot reasoners,” in *Proc. Adv. Neural Inf. Process. Syst.*, 2022, vol. 35, pp. 22199–22213.
- [8] S. Tian et al., “Opportunities and challenges for chatGPT and large language models in biomedicine and health,” *Brief. Bioinf.*, vol. 25, no. 1, 2024, Art. no. bbad493.
- [9] B. Theodorou, C. Xiao, and J. Sun, “Synthesize high-dimensional longitudinal electronic health records via hierarchical autoregressive language model,” *Nature Commun.*, vol. 14, no. 1, 2023, Art. no. 5305.
- [10] K. Singhal et al., “Large language models encode clinical knowledge,” *Nature*, vol. 620, no. 7972, pp. 172–180, 2023.
- [11] U. Katz et al., “GPT versus resident physicians—a benchmark based on official board scores,” *NEJM AI*, vol. 1, no. 5, 2024, Art. no. AIdbp2300192.
- [12] L. Tang et al., “Evaluating large language models on medical evidence summarization,” *npj Digit. Med.*, vol. 6, no. 1, 2023, Art. no. 158.
- [13] Y. Peng, J. F. Rousseau, E. H. Shortliffe, and C. Weng, “AI-generated text may have a role in evidence-based medicine,” *Nature Med.*, vol. 29, no. 7, pp. 1593–1594, 2023.
- [14] A. J. Thirunavukarasu, D. S. J. Ting, K. Elangovan, L. Gutierrez, T. F. Tan, and D. S. W. Ting, “Large language models in medicine,” *Nature Med.*, vol. 29, no. 8, pp. 1930–1940, 2023.
- [15] W. X. Zhao et al., “A survey of large language models,” 2023, *arXiv:2303.18223*.
- [16] P. Sahoo, A. K. Singh, S. Saha, V. Jain, S. Mondal, and A. Chadha, “A systematic survey of prompt engineering in large language models: Techniques and applications,” 2024, *arXiv:2402.07927*.
- [17] J. Li, X. Cheng, W. X. Zhao, J.-Y. Nie, and J.-R. Wen, “Halueval: A large-scale hallucination evaluation benchmark for large language models,” in *Proc. 2023 Conf. Empirical Methods Natural Lang. Process.*, 2023, pp. 6449–6464.
- [18] Y. Tay et al., “UI2: Unifying language learning paradigms,” 2022, *arXiv:2205.05131*.
- [19] Y. Gu et al., “Domain-specific language model pretraining for biomedical natural language processing,” *ACM Trans. Comput. Healthcare (HEALTH)*, vol. 3, no. 1, pp. 1–23, 2021.
- [20] A. Mulyar, O. Uzuner, and B. McInnes, “Mt-clinical bert: Scaling clinical information extraction with multitask learning,” *J. Amer. Med. Informat. Assoc.*, vol. 28, no. 10, pp. 2108–2115, 2021.
- [21] Z. Li, H. Chen, R. Qi, H. Lin, and H. Chen, “DocR-BERT: Document-level R-BERT for chemical-induced disease relation extraction via Gaussian probability distribution,” *IEEE J. Biomed. Health Inform.*, vol. 26, no. 3, pp. 1341–1352, Mar. 2022.
- [22] Y. Hu, V. K. Keloth, K. Raja, Y. Chen, and H. Xu, “Towards precise pico extraction from abstracts of randomized controlled trials using a section-specific learning approach,” *Bioinformatics*, vol. 39, no. 9, 2023, Art. no. btad542.
- [23] G. Zhang, Y. Zhou, Y. Hu, H. Xu, C. Weng, and Y. Peng, “A span-based model for extracting overlapping pico entities from randomized controlled trial publications,” *J. Amer. Med. Inform. Assoc.*, vol. 31, no. 5, pp. 1163–1171, 2024.
- [24] M. Lewis, “Bart: Denoising sequence-to-sequence pre-training for natural language generation, translation, and comprehension,” 2019, *arXiv:1910.13461*.
- [25] B. C. Wallace, S. Saha, F. Soboczenski, and I. J. Marshall, “Generating (factual?) narrative summaries of rcts: Experiments with neural multi-document summarization,” *AMIA Summits Transl. Sci. Proc.*, vol. 2021, 2021, Art. no. 605.
- [26] J. Lee et al., “Biobert: A pre-trained biomedical language representation model for biomedical text mining,” *Bioinformatics*, vol. 36, no. 4, pp. 1234–1240, 2020.
- [27] K. R. Kanakarajan, B. Kundumani, and M. Sankarasubbu, “Bioelectra: Pretrained biomedical text encoder using discriminators,” in *Proc. 20th Workshop Biomed. Lang. Process.*, 2021, pp. 143–154.
- [28] Y. Guo, W. Qiu, Y. Wang, and T. Cohen, “Automated lay language summarization of biomedical scientific reviews,” in *Proc. AAAI Conf. Artif. Intell.*, 2021, vol. 35, pp. 160–168.
- [29] A. Esteva et al., “COVID-19 information retrieval with deep-learning based semantic search, question answering, and abstractive summarization,” *NPJ Digit. Med.*, vol. 4, no. 1, 2021, Art. no. 68.
- [30] B. Le Guellec et al., “Performance of an open-source large language model in extracting information from free-text radiology reports,” *Radiol.: Artif. Intell.*, vol. 6, 2024, Art. no. e230364.
- [31] S. Jia et al., “MedPodGPT: A multilingual audio-augmented large language model for medical research and education,” *medRxiv*, 2024.
- [32] T. Han et al., “Medalpaca—an open-source collection of medical conversational AI models and training data,” 2023, *arXiv:2304.08247*.
- [33] T. Brown et al., “Language models are few-shot learners,” in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, vol. 33, pp. 1877–1901.
- [34] M. Agrawal, S. Hegselmann, H. Lang, Y. Kim, and D. Sontag, “Large language models are few-shot clinical information extractors,” in *Proc. Conf. Empirical Methods Natural Lang. Process.*, 2022, pp. 1998–2022.

- [35] D. V. Veen et al., "Adapted large language models can outperform medical experts in clinical text summarization," *Nature Med.*, vol. 30, no. 4, pp. 1134–1142, 2024.
- [36] J. Wei et al., "Chain-of-thought prompting elicits reasoning in large language models," in *Proc. Adv. Neural Inf. Process. Syst.*, 2022, vol. 35, pp. 24824–24837.
- [37] M. Li, H. Zhou, H. Yang, and R. Zhang, "RT: A retrieving and chain-of-thought framework for few-shot medical named entity recognition," *J. Amer. Med. Inform. Assoc.*, vol. 31, 2024, Art. no. ocae095.
- [38] Y. Hu et al., "Improving large language models for clinical named entity recognition via prompt engineering," *J. Amer. Med. Inform. Assoc.*, vol. 31, 2024, Art. no. ocae259.
- [39] Y.-N. Chuang, R. Tang, X. Jiang, and X. Hu, "Spec: A soft prompt-based calibration on performance variability of large language model in clinical notes summarization," *J. Biomed. Inform.*, vol. 151, 2024, Art. no. 104606.
- [40] Q. Jin, B. Dhingra, Z. Liu, W. Cohen, and X. Lu, "Pubmedqa: A dataset for biomedical research question answering," in *Proc. Conf. Empirical Methods Natural Lang. Process. 9th Int. Joint Conf. Natural Lang. Process.*, 2019, pp. 2567–2577.
- [41] N. Pattisapu, N. Prabhu, S. Bhati, and V. Varma, "Leveraging social media for medical text simplification," in *Proc. 43 rd Int. ACM SIGIR Conf. Res. Develop. Inf. Retrieval*, 2020, pp. 851–860.
- [42] B. Nye et al., "A corpus with multi-level annotations of patients, interventions and outcomes to support language processing for medical literature," in *Proc. Conf. Assoc. Comput. Linguistics. Meeting*, 2018, vol. 2018, Art. no. 197.
- [43] A. Krithara, A. Nentidis, K. Bougiatiotis, and G. Paliouras, "BioASQ-QA: A manually curated corpus for biomedical question answering," *Sci. Data*, vol. 10, no. 1, 2023, Art. no. 170.
- [44] R. Mishra et al., "Text summarization in the biomedical domain: A systematic review of recent research," *J. Biomed. Inform.*, vol. 52, pp. 457–467, 2014.
- [45] C. Wu, W. Lin, X. Zhang, Y. Zhang, W. Xie, and Y. Wang, "PMC-LLaMA: Toward building open-source language models for medicine," *J. Amer. Med. Informat. Assoc.*, 2024, Art. no. ocae045.
- [46] A. Chowdhery et al., "Palm: Scaling language modeling with pathways," *J. Mach. Learn. Res.*, vol. 24, no. 240, pp. 1–113, 2023.
- [47] H. Touvron et al., "LLaMA 2: Open foundation and fine-tuned chat models," 2023, *arXiv:2307.09288*.
- [48] E. Alsentzer et al., "Publicly available clinical bert embeddings," in *Proc. 2nd Clinical Natural Lang. Process. Workshop*, 2019, pp. 72–78.
- [49] S. Wang et al., "GPT-NER: Named entity recognition via large language models," 2023, *arXiv:2304.10428*.
- [50] E. J. Hu et al., "Lora: Low-rank adaptation of large language models," in *Proc. Int. Conf. Learn. Representations*, 2022. [Online]. Available: <https://openreview.net/forum?id=nZeVKeeFYf9>
- [51] P. Lewis et al., "Retrieval-augmented generation for knowledge-intensive NLP tasks," *Adv. Neural Inf. Process. Syst.*, vol. 33, pp. 9459–9474, 2020.